

3/30/2026 Enhanced Sampling

- The lecture will cover
 - importance sampling
 - thermodynamic processes
 - umbrella sampling
 - statistical estimators for thermodynamic properties
- Today's lecture is a key step towards the following learning objective: Explain key concepts related to binding free energy calculations. Compare and contrast molecular docking and binding free energy calculations.
- At the end of this lecture, you should be able to answer the following questions:
 - What is a thermodynamic process?
 - What are the benefits of umbrella sampling?

Importance Sampling

Estimating Thermodynamic Properties

- Molecular simulations are used to calculate thermodynamic and kinetic properties
- In general, the thermodynamic properties are
 - expectation values of an observable $\langle A \rangle = \int A(x)p(x)dx$,including
 - probability of the observable having a certain range of values
 - potential of mean force with respect to the observable
 - free energy differences between thermodynamic states
 - in biomolecular systems, $\Delta G \sim \Delta A$

Importance sampling

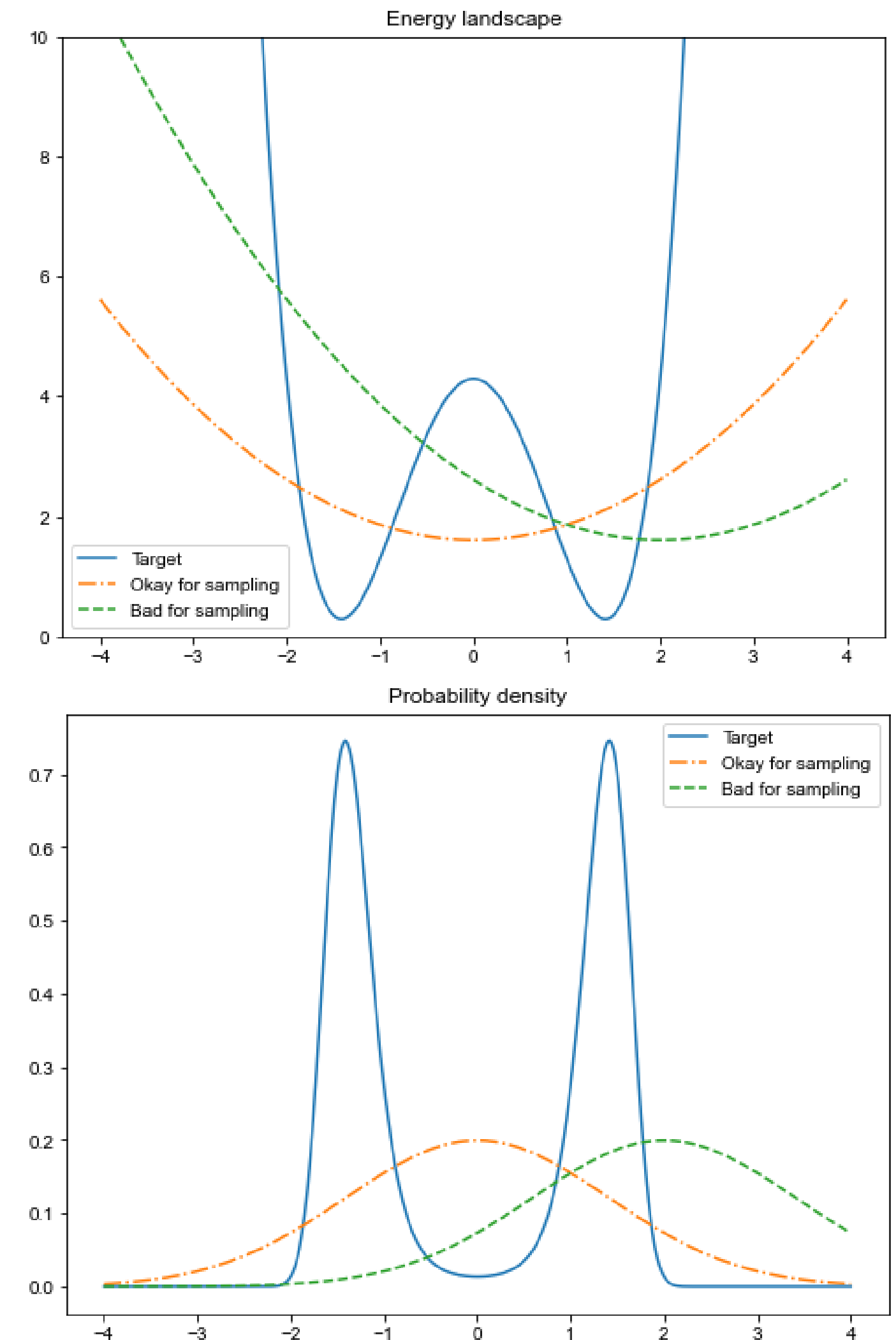
- Sampling from one distribution and estimating quantities in another
- $\langle A \rangle_T = \int A(x) p_T(x) dx = \int A(x) \left(\frac{p_T(x)}{p_S(x)} \right) p_S(x) dx = \langle wA \rangle_S$
- $p_T(x)$ is the probability density in the target distribution
- $p_S(x)$ is the probability density in the sampled distribution
- $w = \left(\frac{p_T}{p_S} \right)$ is the ratio of weights in the two distributions

Why do importance sampling?

- Use one simulation to estimate quantities in multiple thermodynamic states, e.g. different temperatures
- Less computational expense to sample from one distribution, e.g. sample with molecular mechanics and calculate quantities in QM/MM distribution
- Sample from distribution with smaller configuration space, e.g. harmonic restraint towards a crystal structure

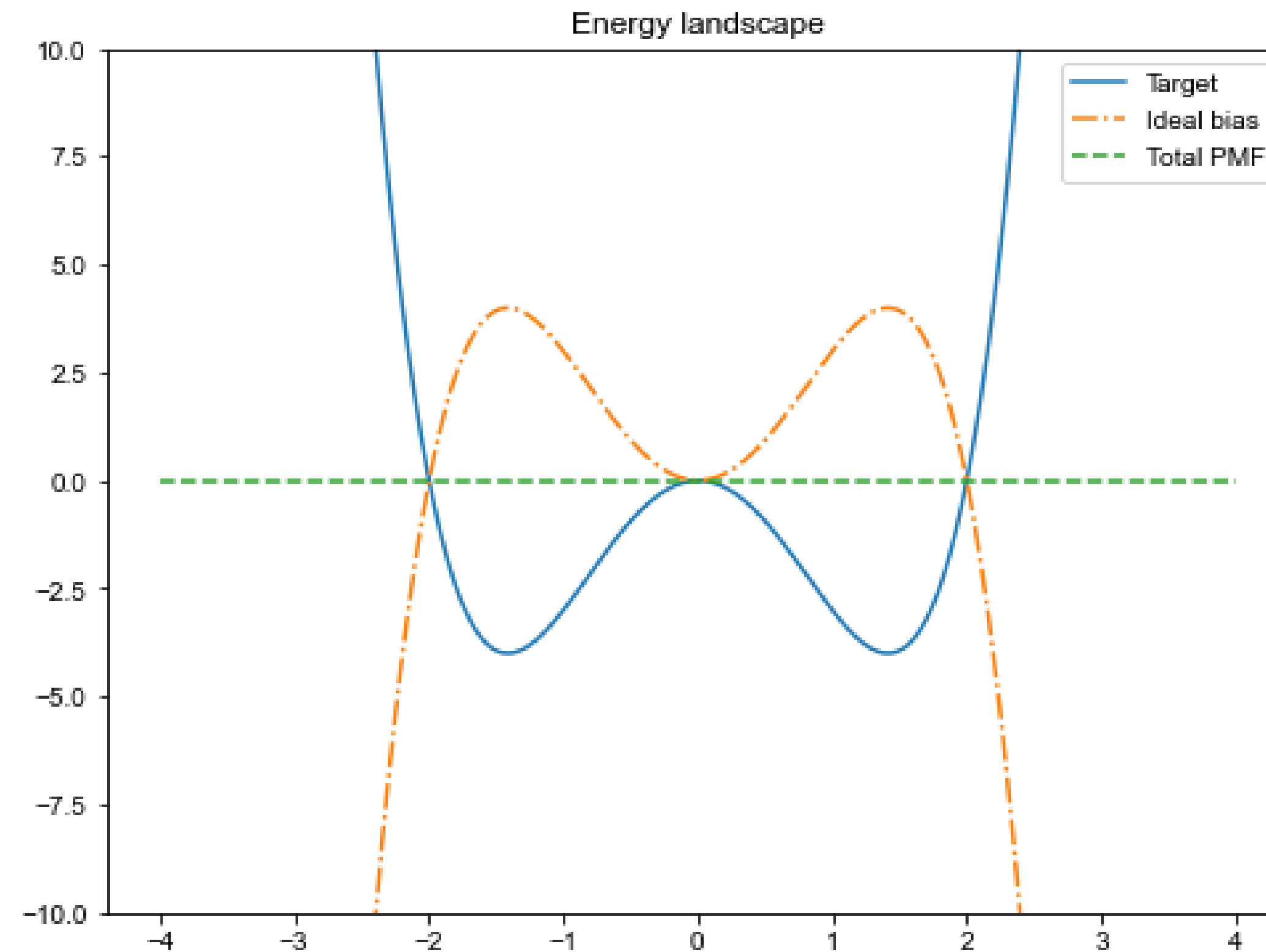
Caveat: Importance sampling

- The target and sampled distribution should be similar
 - If the *support* differs, important configurations may be missed
- If the probability density significantly differs
 - the reweighing term can be noisy
 - thermodynamic expectations will require more samples to converge



Idealized importance sampling

Covers the entire relevant configuration space along an order parameter



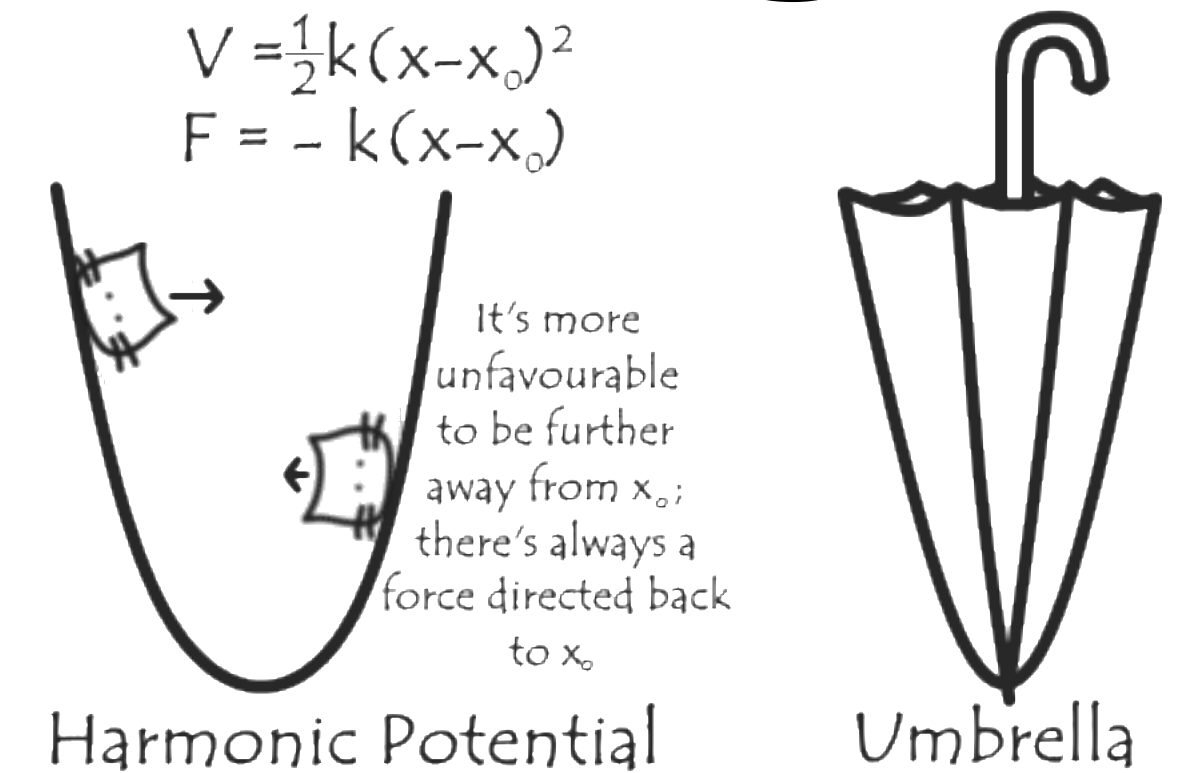
Thermodynamic processes

- A thermodynamic process involves a change in one or more variables that specify a thermodynamic state
- In general and physical chemistry, you learn about specifying a state with
 - temperature
 - pressure/volume
 - number of particles/chemical potential
- Processes include
 - isothermal expansion - a change in volume but not temperature
 - adiabatic expansion - a change in volume without heat transfer in and out of a system
- In molecular simulations, additional variables can define the potential energy
 - harmonic spring constant and center
 - alchemical parameter for
 - transforming one substituent into another
 - decoupling a molecule from solvent

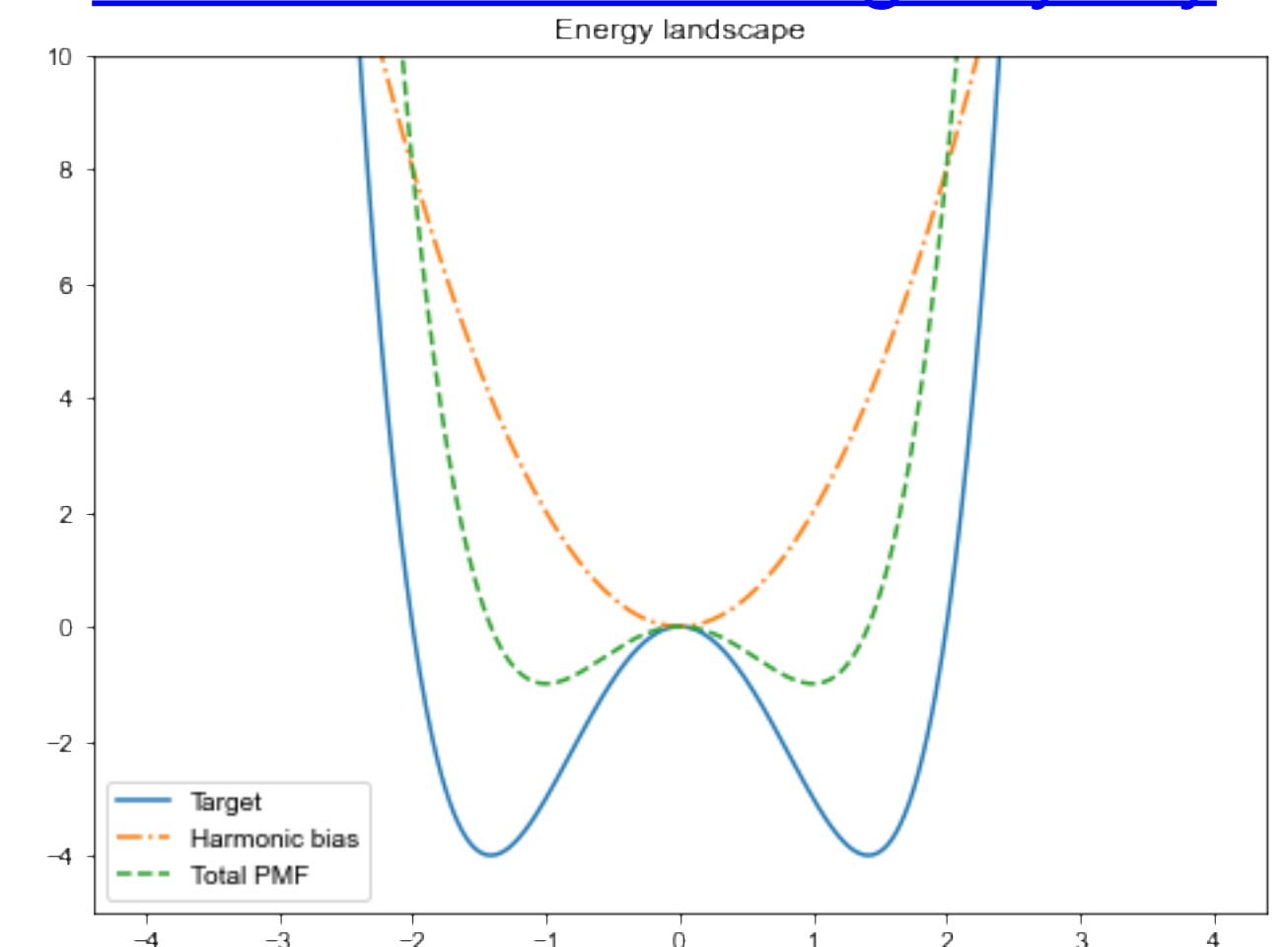
Umbrella Sampling

Harmonic umbrella sampling

- Sampling that *covers* the entire relevant configuration space along an order parameter
- Typically involves
 - a harmonic restraint towards a specific value of the order parameter, $U_b(x) = \frac{1}{2} k (z[x] - z_o)^2$
 - k is the spring constant
 - $z[x]$ is the value of the order parameter for configuration x
 - multiple simulations with different spring centers
 - the observed probability distributions should overlap



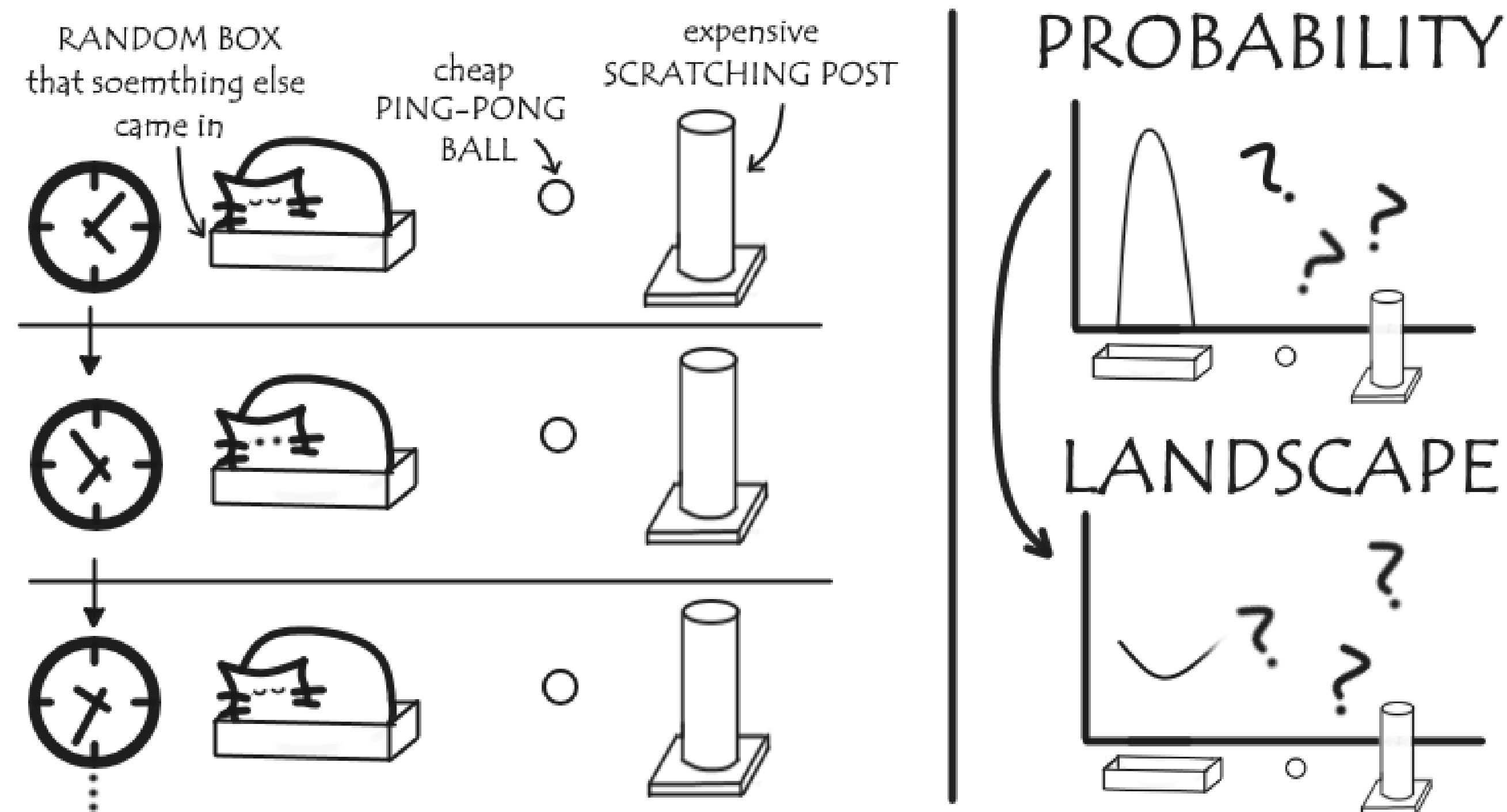
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A cat analogy of umbrella sampling

Which toy does the cat like best?

To quantify this, we can look at the fraction of time they spend near a specific toy.

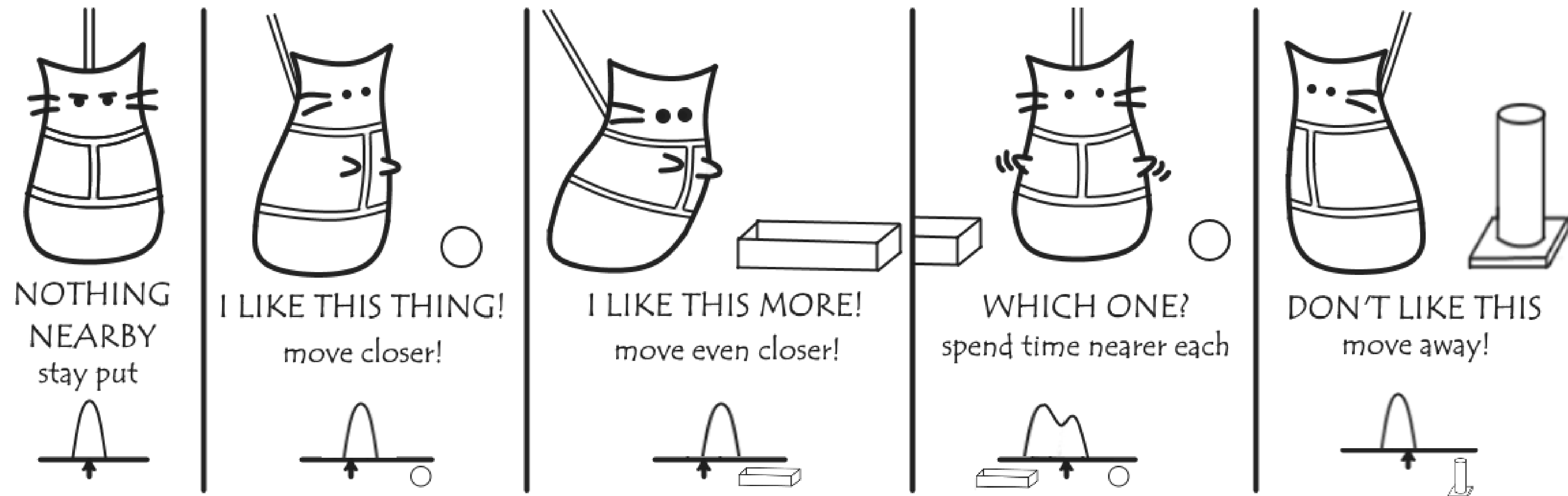


But we can be watching the cat for a long time...

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Umbrella sampling with a cat

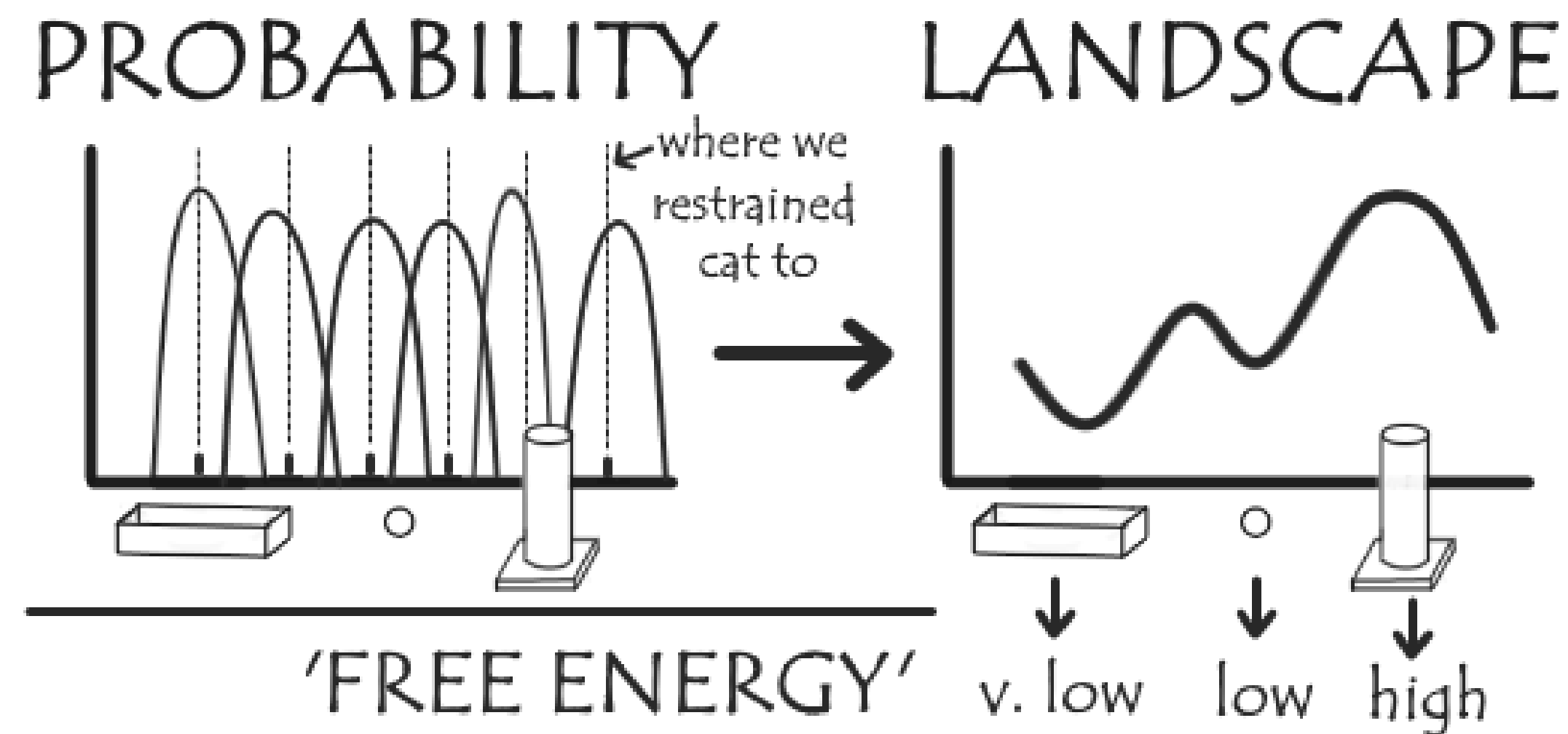
If we put a restraint on the cat, we can determine what they prefer in a smaller area.
Overall, we don't need to watch as long.



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Analysis of umbrella sampling

- Umbrella sampling yields *biased* samples from multiple statistical distributions
- What we usually want is an *unbiased* free energy landscape



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- Samples from individual distributions are reweighed to remove bias
- Samples from multiple distributions are joined
 - to recover a landscape (WHAM, Weighted Histogram Analysis Method)
 - to estimate arbitrary expectation values (MBAR, Multi-state Bennett Acceptance Ratio)

Exercise 9: Umbrella Sampling

Review

- What is a thermodynamic process?
- What are the benefits of umbrella sampling?
- How are umbrella sampling simulations analyzed?